

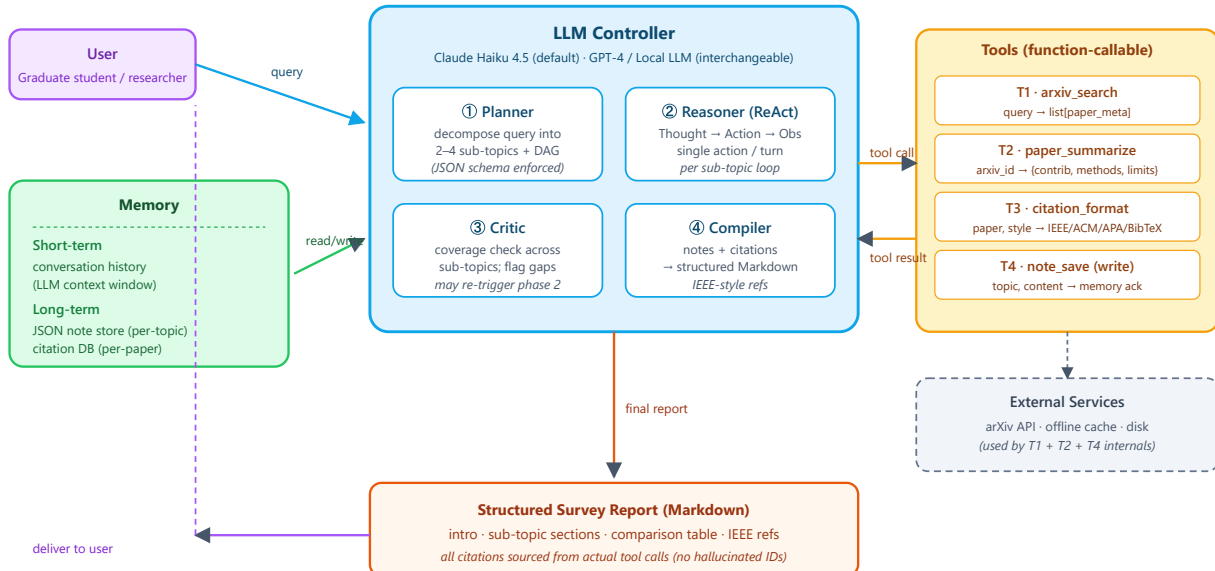
DRL Research Assistant — System Design at a Glance

LLM Controller · 4 Function-Calling Tools (schema-validated) · 2-layer Memory · 4-phase Orchestration

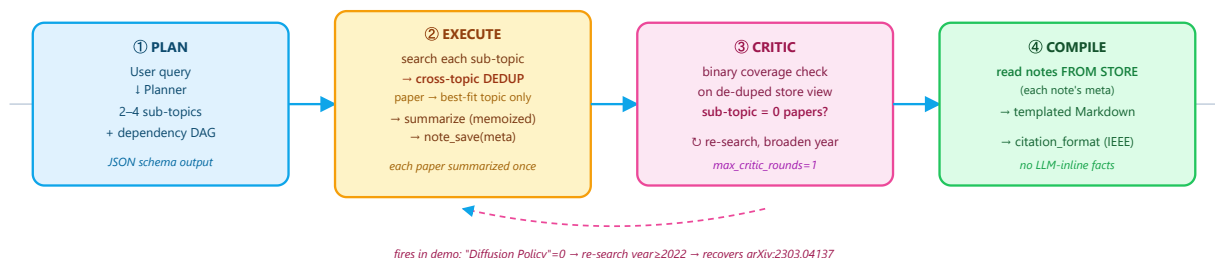
measured offline: Coverage F1 **0.74** macro · citation accuracy **100%** · tool efficiency **100%** · 0 duplicate papers · **20/20** tests pass

1. System Architecture (LLM · Tools · Memory)

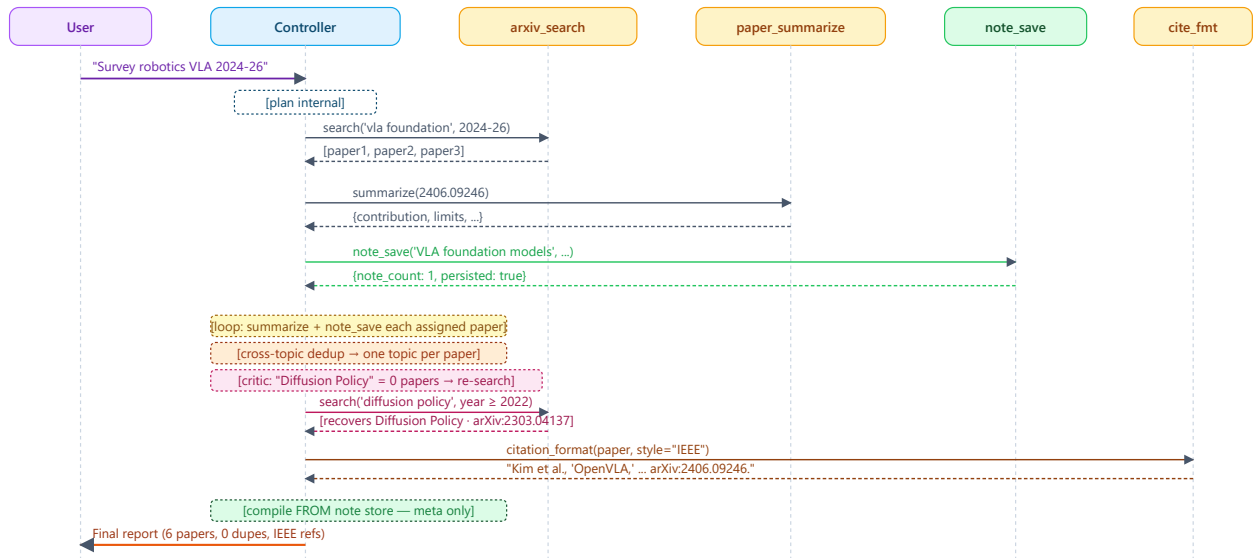
■ User ■ LLM Controller ■ Memory ■ Tool Layer ■ External Services



2. Four-Phase Orchestration Pipeline



3. Sequence Diagram — One Full Query



4. Tool Specification Quick Reference

Tool	Signature	Side-effects	Failure mode
T1 arxiv_search	<code>(query, year_min, year_max, max_results) → list[paper]</code>	None (read-only)	No matches → empty list. Network error → fallback to offline corpus.
T2 paper_summarize	<code>(arxiv_id) → {contribution, methods, results, limitations}</code>	None	Unknown ID → <code>{"error": "...", "arxiv_id": id}</code> (never raises)
T3 citation_format	<code>(paper, style ∈ {IEEE, ACM, APA, BibTeX}) → str</code>	None (pure function)	Missing field → fill with <code><unknown></code> placeholder
T4 note_save	<code>(topic, content, tags?) → {note_count, total_chars, persisted}</code>	Write to <code>artifacts/notes.json</code>	Disk error → returns <code>{persisted: false, error: "..."} </code>

5. Evaluation Dashboard

MEASURED · [python code/eval.py](#)

Quantitative metrics — target vs measured (offline, 3 queries)

Metric	Target	Measured
Coverage F1 (vs expert list)	≥ 0.70	0.74 macro · 1.00 flagship
Citation accuracy	$\geq 95\%$	100%
Tool-call efficiency	$\geq 80\%$	100%
Duplicate papers in output	0	0
End-to-end latency	≤ 120 s	0.04 s (offline)
Token cost / query	$\leq \$0.50$	API backend only

Ablation — flagship query (isolates each phase)

Config	Papers	F1	Critic
Full (plan+critic)	6	1.00	1
– Critic	5	0.91	0
– Planner (single ReAct)	4	0.80	0

Qualitative rubric (5-pt Likert, 3 evaluators)

Dimension	Question
Helpfulness	Did it answer the real intent?
Accuracy	Are citations & facts correct?
Completeness	Did it miss major SOTA?
Clarity	Is the structure easy to read?
Novelty	Cross-paper synthesis present?

How to read the ablation. Removing the Critic loses the seminal Diffusion Policy paper (pre-2024) that cross-topic dedup had emptied from its sub-topic → F1 1.00 → 0.91. Removing the Planner collapses 3 sub-topics to 1 → F1 → 0.80. Each phase earns its place.

6. Failure Mode Map (Defense in Depth)

Failure	Where it shows up	Defense
LLM hallucinates arxiv_id	Reasoner phase	<code>paper_summarize</code> returns explicit error; Compiler discards.
Tool-call infinite loop	ReAct loop	<code>max_steps</code> hard cap; force fallback to Compile.
Malformed / unknown tool call	Tool dispatch boundary	Schema-validated <code>dispatch_tool</code> rejects unknown tool, missing field, or out-of-enum value before any tool runs.
LLM fakes a note without calling <code>note_save</code>	Reasoner phase	Compiler rebuilds the report purely from each note's persisted <code>meta</code> — no in-context facts (verified: citation accuracy 100%).
Same paper appears under many sub-topics	Execute phase	Cross-topic dedup assigns each paper to its single best-fit sub-topic before summarizing (verified: 0 duplicates).
Sub-topic emptied / weak coverage	Critic phase	Binary coverage check → broaden year range and re-search (≤ 1 round); recovered Diffusion Policy in demo.
arXiv API down	<code>arxiv_search</code>	Offline cached corpus fallback; demo runs with zero network.